$-1 \leq \sin(\underline{k}x) \leq 1$

$k \in \mathbb{R}$

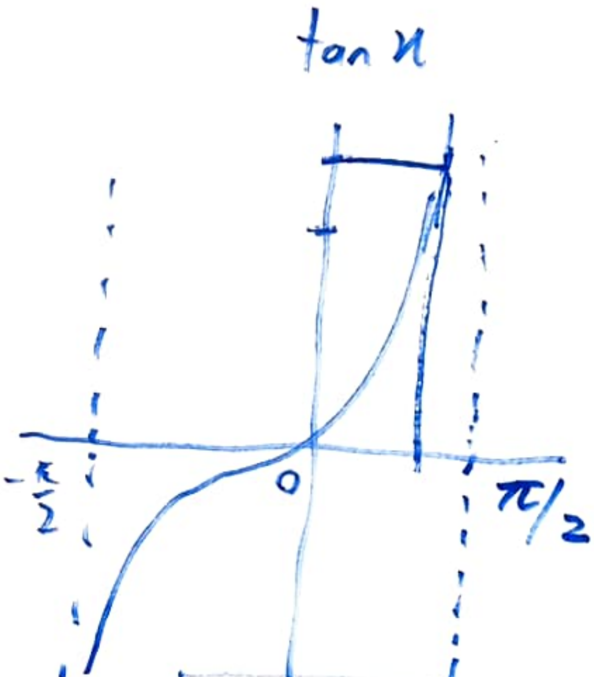
$-1 \leq \cos(\underline{k}x) \leq 1$

$k = \pi$

$-\frac{\pi}{2} < \arctan x < \frac{\pi}{2}$

$0 \leq \operatorname{arcsec} x \leq \pi$

arcsec x

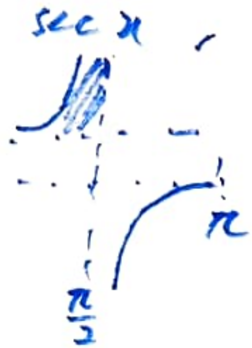


When $x \geq 1$,

$0 \leq \operatorname{arcsec} x < \frac{\pi}{2}$

$\sec x = \frac{1}{\cos x}$

$\longrightarrow$



Ex 9.4 Q 48

$$\sum \frac{\operatorname{arcsec} n}{n^{1.3}}$$

Note that $0 \leqslant \operatorname{arcsec} n < \frac{\pi}{2}$ for $n \geqslant 1$

$$\Rightarrow 0 \leqslant \underbrace{\frac{\operatorname{arcsec} n}{n^{1.3}}}_{\substack{\text{conv.} \\ \text{(by DCT)}}} < \frac{\pi}{2} \cdot \underbrace{\frac{1}{n^{1.3}}}$$

$\sum \frac{1}{n^{1.3}}$ is conv.

$\Downarrow$ $1.3 > 1$

$\frac{\pi}{2} \sum \frac{1}{n^{1.3}}$ is conv.

$\underline{Q20}$ $\sum \frac{1+\cos n}{n^2}$

$-1 \leq \cos n \leq 1$

$\underbrace{0 \leq \frac{1+\cos n}{n^2}}_{\text{conv. by (DCT)}} \leq \underbrace{\frac{2}{n^2}}_{\sum \frac{2}{n^2} \text{ conv.}}$

$\boxed{\coth n \to 1}$ $\left(\coth n = \frac{1}{\tanh n} \to \frac{1}{1} = 1 \right)$

$\boxed{\tanh n \to 1}$ $\left(\tanh n = \frac{\sinh n}{\cosh n} = \frac{\frac{e^{n} - e^{-n}}{\cancel{2}}}{\frac{e^{n}+e^{-n}}{\cancel{2}}} = \frac{e^{n}-e^{-n}}{e^{n}+e^{-n}} = \frac{1-e^{-2n}}{1+e^{-2n}} \to 1 \right)$

$\boxed{n^{\frac{1}{n}} = \sqrt[n]{n} \to 1}$ (Remember it)

$\operatorname{sech} n = \frac{1}{\cosh n} = \frac{2}{e^{x}+e^{-x}} \to 0$

Q52

$$\sum \frac{\boxed{\sqrt[n]{n}}}{n^2} \approx \frac{1}{n^2}$$

Sol. $a_n = \frac{\sqrt[n]{n}}{n^2}$, $b_n = \frac{1}{n^2}$ for $n \geq 1$.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{\frac{\sqrt[n]{n}}{\cancel{n^2}}}{\frac{1}{\cancel{n^2}}} = \lim_{n\to\infty} \sqrt[n]{n} = 1 > 0$$

Since $\sum \frac{1}{n^2}$ conv., $\sum \frac{\sqrt[n]{n}}{n^2}$ conv. by LCT.

Q49 $\sum \frac{\coth n}{n^2} \approx \frac{1}{n^2}$ (LCT)

Let $x \in \mathbb{R}$.

$$\sum \left(\frac{n+\cancel{2026}\,x}{n}\right)^n \Big/ n^2 = \sum \left(1+\frac{x}{n}\right)^n \Big/ n^2 \simeq \frac{1}{n^2}$$

$$\boxed{||\ \text{conv}}$$

4 dir.

$$\boxed{\left(1+\frac{x}{n}\right)^n \to e^x}$$

LCT

$$\lim \frac{a_n}{b_n} \neq 0 \qquad a_n = (\qquad\qquad)$$

$$b_n = \frac{1}{n^2}$$

Div $\to$ Oscillating seq. $\to$ Not work for Ratio Test / Root Test

$\to$ Div. to ∞ / $-\infty$

Ex. 9.5

Q23 $\sum \frac{2+(-1)^n}{(1.25)^n}$

$\left|\pm \frac{1}{1.25}\right| < 1$

$\Rightarrow \sum \left(\frac{1}{1.25}\right)^n$ and $\sum \left(-\frac{1}{1.25}\right)^n$ are convergent.

① Geometric series

$$\sum \frac{2}{(1.25)^n} + \sum \left(\frac{-1}{1.25}\right)^n$$

$$= 2 \sum \left(\frac{1}{1.25}\right)^n + \sum \left(-\frac{1}{1.25}\right)^n$$

② Root Test

$$a_n = \frac{2+(-1)^n}{(1.25)^n} \geq 0$$

$$\sqrt[n]{|a_n|} = \frac{\sqrt[n]{2+(-1)^n}}{1.25}$$

$$1 \leq 2+(-1)^n \leq 3$$

$$\Rightarrow 1 \leq \sqrt[n]{2+(-1)^n} \leq \underline{\sqrt[n]{3}}$$

$$\rightarrow 1$$

By Sqz. Thm., $\sqrt[n]{2+(-1)^n} \rightarrow 1 \Rightarrow \sqrt[n]{|a_n|} \rightarrow \frac{1}{1.25} < 1$

Ex. 9.6 Q26

$$\sum (-1)^{n+1} \sqrt[n]{10}$$

$$a_n = (-1)^{n+1} \sqrt[n]{10}$$

$$|a_n| = \sqrt[n]{10}$$

$\sum |a_n|$ ~~conv~~ div.

$$\lim_{n\to\infty} |a_n| = 1 \neq 0 \Rightarrow \lim_{n\to\infty} a_n \neq 0$$

By n-term test, $\sum a_n$ diverges.